Code No: 155CK **R18**

JAWAHARLAL NEHRU TECHNOLOGICAL UNIVERSITY HYDERABAD
B. Tech III Year I Semester Examinations, January/February - 2023
NATURAL LANGUAGE PROCESSING
(Computer Science and Engineering)

Time: 3 Hours **Max. Marks: 75**

Note: i) Question paper consists of Part A, Part B.
ii) Part A is compulsory, which carries 25 marks. In Part A, Answer all questions.
iii) In Part B, Answer any one question from each unit. Each question carries 10 marks and may have a, b as sub questions.

PART – A

(25 Marks)

1.a)	Define allomorphs.	[2]
b)	What is a lexeme? List its categories.	[3]
c)	What is the use of Treebank?	[2]
d)	Give an example of a phrase structure tree.	[3]
e)	What is the objective of semantic parsing?	[2]
f)	How to identify an event in text?	[3]
g)	List verbal predicates that demand two object arguments.	[2]
h)	Suggest the requirements for a meaning representation system.	[3]
i)	What is meant by cohesion in NLP?	[2]
j)	What is an n-Gram model?	[3]

PART – B

(50 Marks)

2. Describe the following morphological models with illustrations:
a) Dictionary lookup
b) Unification based morphology. [5+5]

OR

3. Make a comparison of generative sequence classification methods and discriminative local classification methods for finding structure of the documents. [10]

4. Elaborate the tokenization and parsing challenges in multilingual content. [10]

OR

5.a) Demonstrate shift-reduce parsing algorithm.
b) Explain the use of probabilistic context free grammars for ambiguity resolution in parsing. [5+5]

6. Describe the methods for word sense disambiguation and the software support for it. [10]

OR

7. Illustrate the methods for resolving entity and event in natural language. [10]

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8. Explain the usage of PropBank in semantic parsing. [10]

OR

9. Provide a detailed description of meaning representation and its need. [10]

10. Make a comparison of syntax based language model and factored language model. [10]

OR

11.a) With an example, discuss demonstrative reference.

b) What is meant by anaphora resolution? Explain the methods used for it. [5+5]

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